Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of a Cloud and AI-Based Universal Non-Destructive Hyperspectral Quality Analysis Technology**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instrument	*Physical Quantity Test/Analysis Instruments*		
Concept of Proposed Technology	**Core Concept** This technology is a universal non-destructive hyperspectral quality analysis platform designed to overcome the limitations of conventional fruit spectroscopy systems that often rely on fruit-specific fixed calibrations/models, whose performance is sensitive to optical variability and requires frequent recalibration under changes in cultivars, seasons, and measurement conditions. Vis-NIR/NIR spectroscopy-based fruit quality assessment and its robustness challenges have been widely reported. (1-3,5) The proposed platform integrates a broadband (Vis-NIR) hyperspectral sensing module with an adaptive deep-learning-based peel/surface correction model to mitigate signal distortions induced by peel properties (e.g., thickness, color, density) and surface effects (e.g., curvature, diffuse reflection, pigment deposition). The feasibility of VNIR hyperspectral imaging for predicting internal quality			

	attributes such as soluble solids content (SSC) in fruits has been demonstrated in the literature. (4-5) In addition, the platform supports cloud-connected model updates, enabling continuous expansion to new fruit species/cultivars without hardware modifications. This approach directly addresses the practical challenge that spectral models may not transfer well across different fruits and conditions, motivating generalized modeling and calibration-transfer strategies. (1,5) **Key Functions and Potential Applications** ● Broadband hyperspectral sensing (Vis-NIR integration): Simultaneously acquires visible (color/freshness) and near-infrared (SSC/acidity/moisture-related) information to support internal quality estimation across diverse peel characteristics. (4-5) ● AI-based peel/surface correction: Performs real-time correction of signal distortions caused by surface curvature, diffuse reflection, and pigment deposition to extract flesh-related signals and improve prediction accuracy. (1-3) ● Cloud-connected model updates: Deploys fruit-optimized models from a cloud server and expands coverage as global data are accumulated, mitigating poor cross-fruit transferability. (1,5) ● Application areas: Orchard/packing-house grading, distribution quality management, export quality certification, smart farming, and automated sorting systems. (1-3)
Rationale for Joint R&D Project	**Critical Technical Barriers** Conventional non-destructive fruit quality inspection technologies typically rely on fixed, fruit-specific calibration/models. However, they inherently suffer from persistent Optical Uncertainty, where hyperspectral/Vis-NIR signals are distorted by optical interferences-scattering, absorption, and diffuse reflection-driven by variations in peel morphology such as thickness, density, and color. This leads to substantial performance degradation under changes in origin, cultivar, season, and measurement environment, and often necessitates repeated recalibration, creating significant time and cost burdens that hinder field-scale deployment and global scalability. These robustness limitations and practical constraints have been widely discussed in the fruit/produce spectroscopy and hyperspectral imaging literature. (1-2,5) **Why Global Collaboration?** This joint R&D goes beyond simply collecting more data. It aims to structurally overcome the above barriers by integrating physics-based theory (Physics) with data-driven intelligence (AI). ● Advancing foundational models via Physics-Informed AI (Purdue University): By leveraging Purdue University's optical simulation capabilities and agri-bioscience testing infrastructure, the project will physically model peel-induced optical interference and incorporate this knowledge into the learning process through Physics-Informed AI, thereby enhancing robustness against data bias and environmental variability. The physics-informed learning paradigm has been established as an effective approach for improving generalization under uncertainty and limited or biased observations. (8)

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	● Addressing global domain shift and achieving universal generalization: Spectral models often exhibit limited transferability when deployed across different fruits and operating conditions, motivating strategies such as calibration transfer/standardization and model updating/transfer learning to mitigate domain shift. (1,3,5) Domestic datasets alone are insufficient to capture the diversity of overseas specialty fruits and North American market/environmental factors. By utilizing Purdue's local networks and testbeds to secure and validate diverse datasets (across fruits, origins, and seasons), the project will finalize a universal generalization framework enabling multi-fruit analysis on a single device. (1,3,5) **Expected Impact** Through collaboration with the overseas research partner, the project will establish: ● international-level quality analysis protocols and benchmarking/evaluation methodologies, ● field validation of performance and reliability in overseas environments, and ● global applicability and credibility of a SaaS-based model update platform. Ultimately, this will accelerate commercialization and enable the Korean SME to build a strong competitive position and entry barriers in the global non-destructive quality analysis market. (1-3,5)
Key Words	1. Non-destructive Quality Inspection 2. Hyperspectral Imaging 3. AI-driven Analysis 4. Cloud-based Model Update 5. Universal / Multi-fruit Generalization

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	**R&D Objectives** To enable non-destructive analysis of internal quality indicators (e.g., soluble solids content/Brix, moisture- and acidity-related indices) across diverse fruit species using a single portable device, this project aims to develop, validate, and establish a commercialization basis for a universal non-destructive quality analysis system that integrates: ● an ultra-compact broadband (Vis-NIR) spectroscopic module, ● an adaptive (variable) deep-learning-based peel/surface correction and quality prediction algorithm, and ● a cloud-connected model update and data management platform, thereby minimizing hardware dependency and ensuring scalable expansion to new fruits and cultivars. **Detailed Objectives** 1) Ultra-compact Precision Optical Hardware ● Design and develop an ultra-compact broadband (Vis-NIR) spectroscopic module capable of robust signal acquisition across diverse peel characteristics (thickness/color/density). ● Develop impact-resistant and field-ready optical packaging and mechanical design that maintains optical-axis stability under harsh on-site conditions (e.g., packing facilities and distribution environments). 2) Adaptive AI Algorithms ● Develop an adaptive deep-learning-based peel/surface correction and quality prediction algorithm that learns and compensates for peel/surface-induced signal distortions in real time. ● Implement model generalization and update strategies (e.g., transfer learning) to support multi-fruit applicability and continuous performance improvement. 3) Global Quality Data & Cloud Platform ● Build a cloud-based data and model update system that enables global collection/management of new fruit/cultivar datasets and supports model selection/download to the device. ● Establish standardized protocols for data acquisition, labeling, and validation to ensure reliability, scalability, and global deployability of the universal model.			
R&D Target Specifications	● SSC/Brix prediction performance: Coefficient of determination $R^2 \geq 0.95$, with an error range within ±0.5 Brix ● Measurement repeatability: Relative standard deviation (RSD) ≤ 5% ● Number of fruit species supported: ≥ 30 species ● Measurement time: ≤ 2 s per sample			

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	● AI inference time: ≤ 0.5 s per sample (on-device computation) ● Environmental robustness: Splash-resistant for daily use; operating temperature 5-40°C ● Operating environments: Orchards, packing facilities (sorting/packing houses), and distribution sites
R&D Plan	Year 1: Securing Core Foundational Technologies ● Design a portable optical engine module integrating a high-power light source and MEMS sensor for broadband (Vis-NIR) spectroscopy, and build a first bench-level prototype. ● Implement baseline spectral preprocessing and an initial deep-learning-based peel/surface correction model, and set up an evaluation environment. ● Build a pilot domestic dataset and conduct initial calibration and repeatability tests. Year 2: Platform Development and Data Expansion ● Improve the universal AI model using the expanded datasets and implement a transfer-learning pipeline for rapid onboarding of new fruits. ● Produce a field-test prototype (Rev.B), including mechanical/packaging design, and conduct performance tuning, calibration, and optimization. ● Develop an MVP of a cloud-connected data/model management system, including model updates and version control. Year 3: Reliability Validation and Commercialization ● Finalize product requirements through domestic and overseas demonstrations, verifying durability, accuracy, and repeatability. ● Obtain country-specific certifications (e.g., radio/wireless and safety), design mass-production tooling, and manufacture the final product. ● Deploy the production-grade SaaS model-update platform and establish global sales/partner networks. ● Secure official test reports from accredited certification/testing bodies and launch commercialization.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Advancing foundational technologies to reduce optical uncertainty and developing a globally generalizable AI model ● Investigate and model the optical scattering and absorption mechanisms induced by fruit peel properties (thickness/color/density) and surface characteristics, leveraging simulation- and analysis-based approaches. ● Design and enhance universal deep-learning / physics-informed AI algorithms that are robust to domain shifts (origin/season/cultivar/measurement environment), with a focus on improving generalization performance. ● Build and provide a global spectral database centered on overseas specialty fruits (e.g., tropical fruits), and establish/provide standardized protocols for data acquisition, labeling, and reference measurements. ● Conduct independent performance validation and benchmarking using overseas testbeds, and provide scientific and technical verification outputs (e.g., evaluation reports and validated data packages). ● Define and prioritize target overseas fruits/cultivars and support international deployment and certification requirements.
Expected R&R of Korean SME Partner	Leading the development of portable hyperspectral hardware and commercialization of a cloud-connected platform ● Design, build, and industrialize portable hyperspectral hardware (optics/light source and sensor, electronics, and mechanical structure), including environmental packaging, durability/reliability improvements, and field usability (UX) considerations for mass-production readiness. ● Develop and operate a user-oriented mobile application and cloud platform, enabling model selection/download, model version management and deployment, data upload/management, and performance logging/monitoring. ● Conduct domestic dataset collection, reference-measurement-based labeling, and field testing in compliance with standardized protocols; perform calibration/performance tuning and incorporate user feedback from demonstrations. ● Lead country-specific certifications and commercialization, including radio/wireless and safety approvals, securing accredited test reports, establishing manufacturing readiness, and expanding domestic and global sales/partnership networks.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical Outcomes: Establishing a globally deployable hyperspectral AI analytics framework and standardization basis ● Secure adaptive (universal) spectral analytics algorithms robust to variations in fruit species and operating conditions, together with a cloud-connected model update (SaaS) framework. ● Build a global hyperspectral database covering overseas specialty fruits (e.g., tropical fruits), including: raw spectra, reference-measurement labels (e.g., SSC/Brix), and metadata such as origin, cultivar, season, and measurement conditions. ● Establish and validate standardized protocols for data acquisition/labeling and benchmarking/evaluation methodologies, supported by joint demonstration results and verification reports. ● File core technology patents (domestic and international, including PCT) and secure country-specific certifications and accredited test reports (e.g., radio/wireless, safety, and performance). Economic/Industrial Impact: Commercialization of an export-oriented product/platform and scalable industrial applicability ● Enhance global export competitiveness (e.g., Southeast Asia and Europe) through a multi-fruit, cloud-updatable solution applicable to grading and distribution/packing operations. ● Secure a scalable core technology stack (sensing + AI + platform) extendable to smart agriculture (smart farming) and automated sorting/inspection systems. ● Import substitution effect: Replace reliance on imported Brix meters (e.g., Atago, Japan) with a Korean-made portable device and platform, reducing equipment acquisition and operational costs. Societal Impact: Reduced fresh-produce waste and improved farm income stability ● Reduce fresh-produce waste and associated carbon emissions by enabling early screening of fruits with internal quality degradation that is not easily detectable by appearance during distribution. ● Improve pricing fairness and transparency through objective quality data, enabling high-quality producers to receive appropriate value, strengthening farmer income stability and consumer trust.

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